

Seeing a Conjugate Point

A guided tour of the second-order theory of the calculus of variations,
with an interactive explorer

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Abstract

An extremal of a variational problem — a solution of the Euler–Lagrange equation — is only a *candidate* for a minimum. Whether it really is one is decided by the second variation, and the second variation is decided, almost entirely, by one geometric event: whether the extremals that leave the starting point alongside ours come back and meet it again before the end of the interval. The first such meeting point is the *conjugate point*.

This guide builds that idea in three layers (a picture, the reasoning, the equations) and then hands you an interactive explorer, `conjugate_point_explorer.m`, in which you type a Lagrangian and *watch* it happen. Four independent views of the same fact appear side by side: neighbouring extremals crossing, the zero of the Jacobi field, the slope of the shooting function, and a negative eigenvalue of the second variation — with the cost J actually going down along the direction that eigenvalue points to. Every number quoted in the experiments was computed by `conjugate_point_study.m` and is checked there against a closed form or a second, independent instrument.

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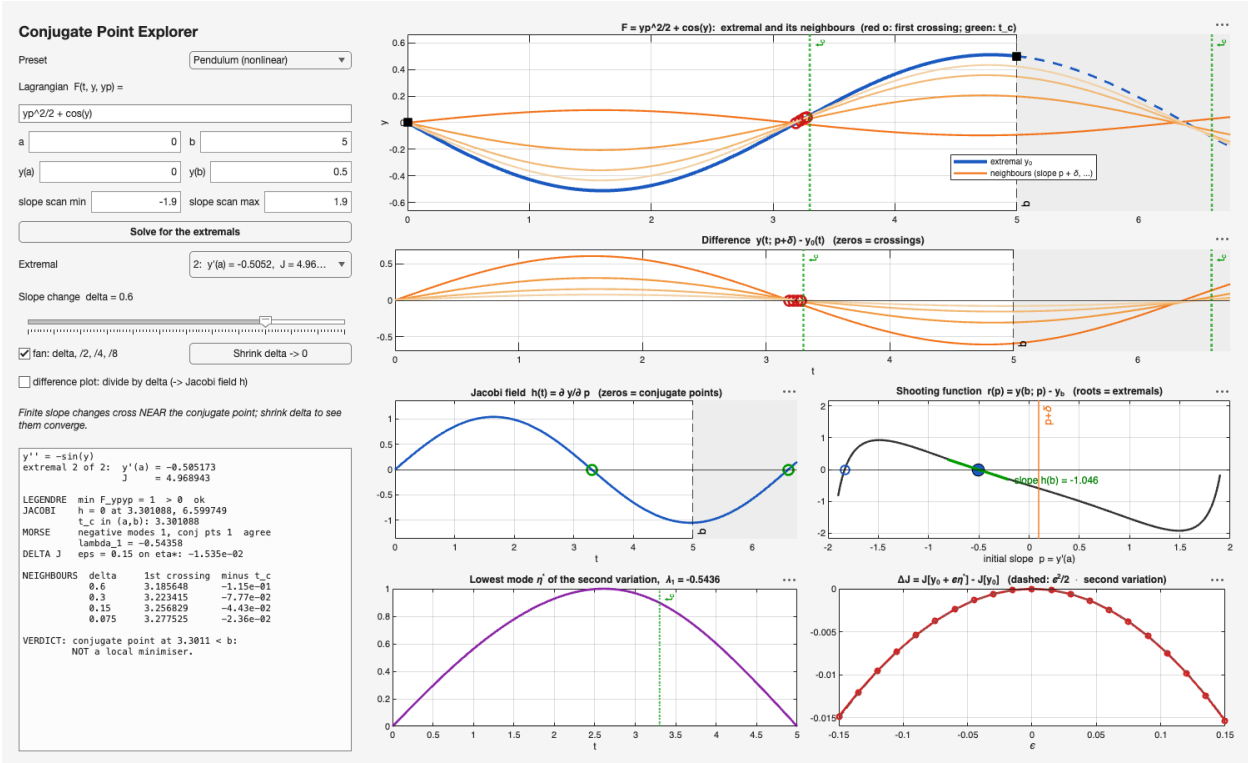


Figure 1: The explorer on the pendulum preset. Top: the extremal (blue) and four neighbours leaving the same start point with slightly different slopes (orange); they cross the extremal near t_c (green), well before $b = 5$. Second strip: the difference $y(t; p + \delta) - y_0(t)$, whose zeros are the crossings. Third row: the Jacobi field (left) and the shooting function (right). Bottom: the lowest mode of the second variation (left), and the cost change along it (right) — a downward parabola, so this extremal is *not* a minimiser.

1 What you will see

Run `conjugate_point_explorer` from the folder. Pick a preset or type your own $F(t, y, y')$ (write y' as `yp`), the interval $[a, b]$ and the end values, and press *Solve*. The explorer finds *every* extremal in the slope range you give it, and for the one you select it draws the six panels of Figure 1 and writes a readout with a verdict. The slider changes the slope perturbation δ ; *Shrink* animates $\delta \rightarrow 0$. The checkbox *divide by delta* rescales the difference strip to $(y(t; p + \delta) - y_0(t))/\delta$, which is the quantity that becomes the Jacobi field h as $\delta \rightarrow 0$ (drawn dashed beside it).

2 The problem, and why an extremal is only a candidate

ELI5

You want the cheapest path from one point to another, where “cheap” is measured along the whole path. The Euler–Lagrange equation finds paths where no *tiny* nudge changes the cost at first glance — the bottom of a valley, but also a mountain pass, or the top of a hill. To know which one you are standing on, you have to look at how the cost *curves* as you move away.

Rigor

The simplest problem of the calculus of variations:

$$\text{minimise } J[y] = \int_a^b F(t, y(t), y'(t)) dt, \quad y(a) = y_a, \quad y(b) = y_b.$$

For $y = y_0 + \varepsilon \eta$ with $\eta(a) = \eta(b) = 0$,

$$J[y_0 + \varepsilon \eta] = J[y_0] + \varepsilon \delta J[\eta] + \frac{\varepsilon^2}{2} \delta^2 J[\eta] + O(\varepsilon^3).$$

$\delta J = 0$ for all η gives the Euler–Lagrange equation $\frac{d}{dt} F_{y'} = F_y$; its solutions are *extremals*. Expanded and solved for y'' (this is what `cov.problem.m` does symbolically),

$$y'' = g(t, y, y') = \frac{F_y - F_{y't} - F_{y'y} y'}{F_{y'y'}},$$

which needs $F_{y'y'} \neq 0$. The second variation is

$$\delta^2 J[\eta] = \int_a^b (P \eta'^2 + 2R \eta \eta' + Q_0 \eta^2) dt, \quad P = F_{y'y'}, \quad R = F_{y'y}, \quad Q_0 = F_{yy},$$

all evaluated along y_0 . Integrating the middle term by parts ($2R \eta \eta' = R(\eta^2)'$, and η vanishes at both ends) gives the form used in the theory,

$$\delta^2 J[\eta] = \int_a^b (P \eta'^2 + Q \eta^2) dt, \quad Q = F_{yy} - \frac{d}{dt} F_{yy'}.$$

A minimum needs $\delta^2 J \geq 0$ for every admissible η .

Intuition

The η'^2 term dominates for fast wiggles: a variation that oscillates rapidly has a huge derivative and a small amplitude. So if $P < 0$ anywhere, wiggling there drives $\delta^2 J$ negative and the extremal is not a minimum. That is **Legendre’s necessary condition**, $P = F_{y'y'} \geq 0$. But $P > 0$ is not enough: slow, long variations feel Q , and if Q is negative over a long enough stretch it can win. *How long is long enough* is exactly the question the conjugate point answers.

3 Neighbours, the Jacobi field, and the conjugate point

ELI5

Throw a ball, then throw a second one from the same spot at a very slightly steeper angle. The two paths separate — and on some problems they come back together later. Where they meet again is special: from there on, there are two nearly equal ways to have got there, and the original path stops being the best.

Rigor

Let $y(t; p)$ be the extremal with $y(a) = y_a$, $y'(a) = p$. Differentiate the Euler–Lagrange equation with respect to p and put $h = \partial y / \partial p$:

$$\frac{d}{dt}(R h + P h') - (Q_0 h + R h') = 0 \implies \frac{d}{dt}(P h') - Q h = 0, \quad h(a) = 0, \quad h'(a) = 1.$$

This is the **Jacobi equation**. It is the Euler–Lagrange equation of the quadratic functional $\delta^2 J$ itself (the “accessory problem”), and it is the linearisation of $y'' = g$: $h'' = g_y h + g_{y'} h'$, which is the form `cov_shoot.m` integrates alongside the extremal.

A point $c \in (a, b]$ is **conjugate to a** if $h(c) = 0$.

Intuition

The neighbour with slope $p + \delta$ is

$$y(t; p + \delta) = y_0(t) + \delta h(t) + O(\delta^2).$$

So where h has a zero, the neighbour — to first order — is back on the extremal. The neighbours *refocus*. On a linear problem (a quadratic F) the $O(\delta^2)$ term is zero and every neighbour crosses at exactly the conjugate point, whatever δ (Figure 2). On a nonlinear problem a finite δ crosses *near* it, and the crossing converges to t_c as $\delta \rightarrow 0$ (Section 6).

4 The theorem

Theorem 1 (Jacobi; see [1, 2]). *Let y_0 be an extremal with $P = F_{y'y'} > 0$ on $[a, b]$.*

1. (Necessary) *If y_0 is a weak local minimiser, there is no point conjugate to a in (a, b) .*
2. (Sufficient) *If there is no point conjugate to a in $(a, b]$, y_0 is a strict weak local minimiser.*

Rigor

Why a conjugate point inside kills the minimum. Suppose $h(c) = 0$ with $a < c < b$. Build the broken variation

$$\hat{\eta}(t) = \begin{cases} h(t), & a \leq t \leq c, \\ 0, & c < t \leq b. \end{cases}$$

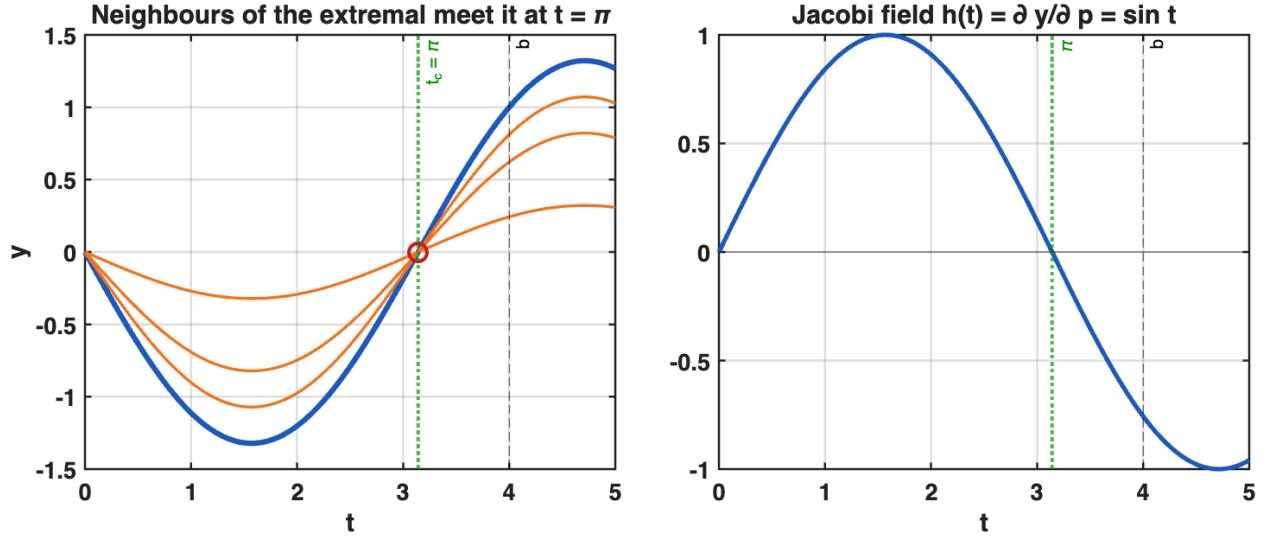


Figure 2: $F = y'^2 - y^2$ on $[0, 4]$: the extremal $y_0 = \sin t / \sin 4$ and its neighbours with $\delta = 1, 0.5, 0.25$ all pass through $(\pi, 0)$, because the Jacobi field is $h = \sin t$ (right). $\pi < b = 4$: not a minimiser.

It is admissible (continuous, zero at both ends) and not identically zero, and

$$\delta^2 J[\hat{\eta}] = \int_a^c (Ph'^2 + Qh^2) dt = [Phh']_a^c - \int_a^c h((Ph')' - Qh) dt = 0,$$

since $h(a) = h(c) = 0$ and h solves the Jacobi equation. If y_0 were a minimiser, $\delta^2 J \geq 0$ everywhere, so $\hat{\eta}$ would *minimise* $\delta^2 J$ and would have to satisfy the corner (Weierstrass–Erdmann) condition of that problem: $P\hat{\eta}'$ continuous at c . From the right $\hat{\eta}' = 0$; from the left it is $h'(c)$, so $Ph'(c) = 0$, hence $h'(c) = 0$. Together with $h(c) = 0$ that forces $h \equiv 0$ by uniqueness — but $h'(a) = 1$. Contradiction: $\delta^2 J$ takes negative values, and y_0 is not a minimum. Rounding off $\hat{\eta}$'s corner produces such a negative direction explicitly.

Counting (Morse). More is true: the number of negative eigenvalues of $\delta^2 J$ on variations vanishing at a and b equals the number of points conjugate to a in (a, b) [3]. `cov_second_variation.m` discretises $\delta^2 J$ with finite elements and counts them — an instrument that never solves the Jacobi equation, so its agreement with h 's zeros is genuine evidence.

Where it breaks / what is assumed

- **Weak, not strong.** The theorem compares y_0 with curves close in value *and* slope. Against curves that are close in value but may have very different slopes (a strong minimum), one needs in addition the Weierstrass excess function and a field of extremals [1]. The explorer tests weak minimality only.
- **Legendre first.** Everything assumes $P > 0$. If $P < 0$ somewhere, Legendre's necessary condition already fails and the extremal is not a minimum (the explorer says so and marks the Morse count as not applicable); if $P = 0$ somewhere, these tests are inconclusive.

- **A conjugate point exactly at b** is the borderline case: the second variation has a zero direction and the test is inconclusive.
- **Numerics.** Extremals are found by scanning a slope range you choose; an extremal outside it is not found. Zeros of h are located by event detection at `RelTol` 10^{-10} ; the eigenvalue count uses 300–400 finite elements.

5 Four views of one fact

Intuition

The explorer puts four independent views of the conjugate point on screen at once. They are computed by different code, so when they agree you are seeing the theory, not an artefact:

1. **Neighbours cross.** Fly the extremal with slope $p + \delta$ and find where it crosses y_0 (`cov_perturbed.m`). The difference strip shows the same event as a zero of $y(t; p + \delta) - y_0(t)$; divided by δ that difference converges to h , which is how views 1 and 2 become one.
2. **h vanishes.** Integrate the Jacobi equation and find its zeros (`cov_shoot.m`).
3. **The shooting function turns.** The extremals are the roots of $r(p) = y(b; p) - y_b$, and $r'(p) = h(b)$. If b itself is conjugate, $r'(p) = 0$ at the root: two extremals are about to merge or be born. That is a fold (Experiment 5).
4. **The second variation goes negative.** The number of negative eigenvalues of $\delta^2 J$ equals the number of conjugate points in (a, b) , and the lowest eigenvector η^* is a direction in which J actually decreases: $J[y_0 + \varepsilon \eta^*] - J[y_0] < 0$, computed exactly, not just predicted (`cov_second_variation.m`).

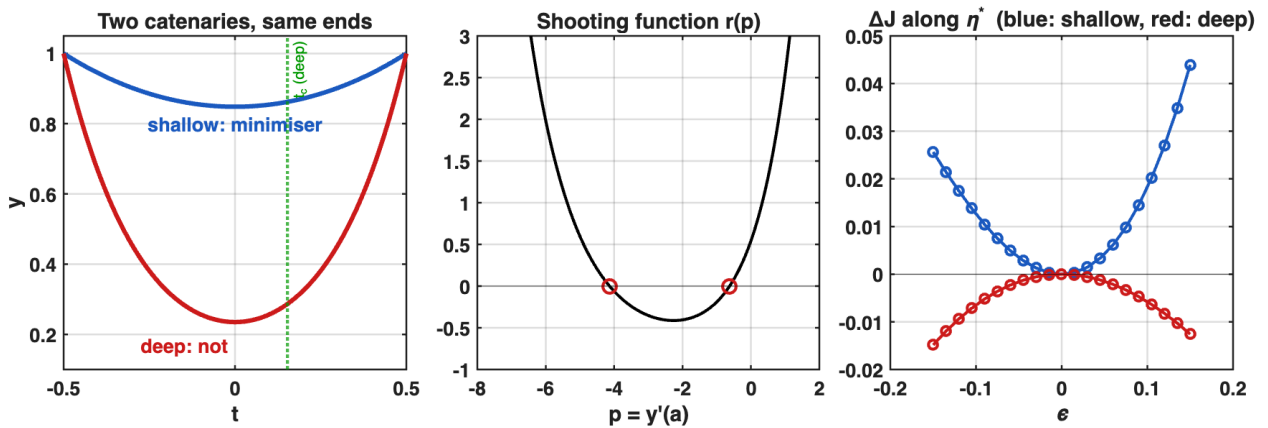


Figure 3: The minimal surface of revolution, $F = y\sqrt{1+y'^2}$, $y(\pm\frac{1}{2}) = 1$. Left: two catenaries join the ends; the deep one has a conjugate point at $t_c = 0.1529$. Middle: the shooting function has two roots, with slopes of opposite sign ($r' = h(b)$). Right: ΔJ along each extremal's lowest mode — up for the shallow one, down for the deep one.

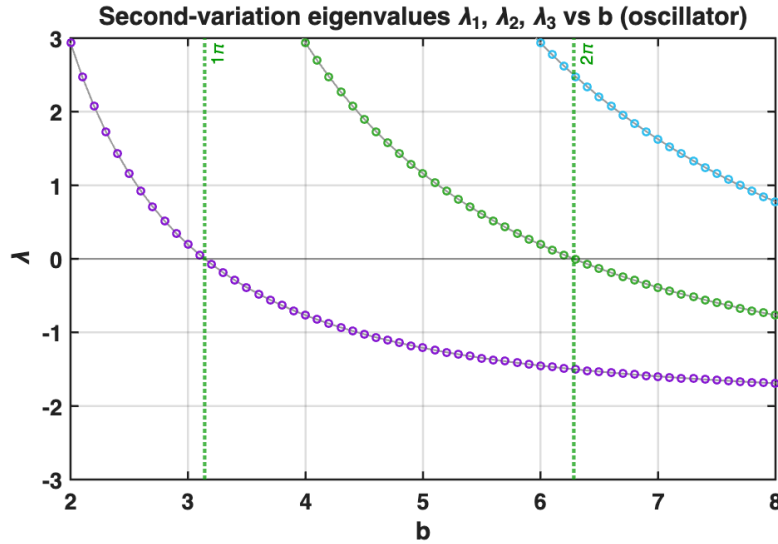


Figure 4: The three lowest eigenvalues of the second variation of $F = y'^2 - y^2$ as the interval $[0, b]$ grows (circles: finite elements; grey: the closed form $2((k\pi/b)^2 - 1)$). Each crosses zero exactly as b passes a conjugate point $k\pi$: the Morse count in one picture.

6 A finite nudge versus the limit

Intuition

The definition of a conjugate point is a *limit*: the crossing point of neighbours as $\delta \rightarrow 0$. With a finite δ on a nonlinear problem, the neighbour crosses somewhat early or late, off by $O(\delta)$. This is why the explorer draws a fan $(\delta, \delta/2, \delta/4, \delta/8)$ and has a *Shrink* button: the crossings march toward t_c , and the readout prints their distance from it.

7 Guided experiments

Open the explorer (`conjugate_point_explorer`) and work through these in order. Each asks you to *predict* first. The checkpoints are the values `conjugate_point_study.m` computed and verified.

Experiment 1: the oscillator, before and after π

Preset *Oscillator*, $b = 3$. Predict: $F = y'^2 - y^2$ gives $y'' = -y$, so $h = \sin t$. Where are the conjugate points? Is $b = 3$ before or after the first one? Read the verdict. Switch to *Oscillator*, $b = 4$ and compare.

Checkpoint

$b = 3$: one extremal, $J = \cot 3 = -7.015253$, $\lambda_1 = +0.1933$, verdict *weak local minimiser*. $b = 4$: $y'(0) = 1/\sin 4 = -1.32134871$, $J = \cot 4 = 0.863691$, $t_c = \pi$ to ten digits, $\lambda_1 = -0.7663$, every neighbour crosses at π for any δ , verdict *not a minimiser*.

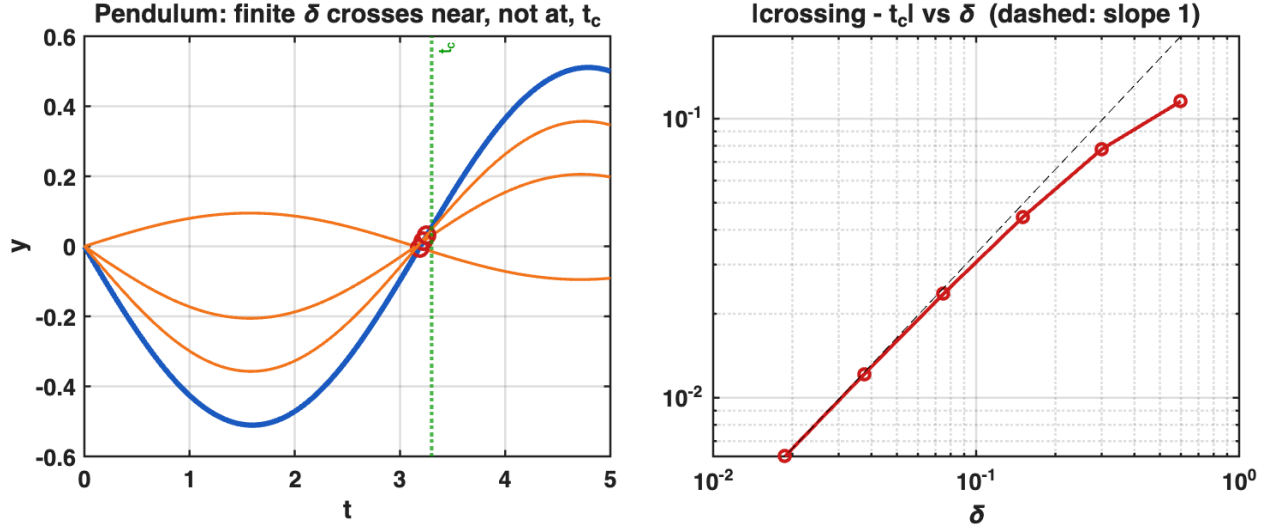


Figure 5: The pendulum $F = y'^2/2 + \cos y$ on $[0, 5]$. Left: neighbours with $\delta = 0.6, 0.3, 0.15$ cross near, not at, $t_c = 3.3011$. Right: the crossing error halves when δ halves (slope 1 on log-log axes).

Experiment 2: two conjugate points

Preset *Oscillator*, $b = 7$. Predict how many negative modes the second variation has, then look at the Morse line.

Checkpoint

Zeros of h at π and 2π inside $(0, 7)$; negative modes = 2; $\lambda_1 = -1.5972$. Figure 4 shows why: one eigenvalue crosses zero each time b passes $k\pi$.

Experiment 3: a finite nudge on a nonlinear problem

Preset *Pendulum*. There are two extremals; select the second. Compare the first crossings of the four neighbours with t_c in the readout, then press *Shrink* and watch them converge. Now select the first extremal: where is its conjugate point?

Checkpoint

Extremal 2: $y'(0) = -0.505173$, $J = 4.968943$, $t_c = 3.301088$. Crossing errors at $\delta = 0.6, 0.3, 0.15, 0.075, 0.0375, 0.01875$: 0.115, 0.078, 0.044, 0.024, 0.012, 0.0062 (halving ratios rise to 1.97: first order). Now tick *divide by delta*: the four rescaled differences lie close to the dashed Jacobi field and approach it as δ shrinks (their largest distance from h on $[0, 5]$ is 0.116, 0.078, 0.045, 0.024 for $\delta = 0.6, 0.3, 0.15, 0.075$). On the oscillator presets they sit exactly on it, for any δ . Extremal 1: $y'(0) = -1.834052$, $J = 3.690547$, first zero of h at $6.636 > b$: a weak local minimiser, and the lower of the two costs.

Experiment 4: two catenaries

Preset *Minimal surface*. $F = y\sqrt{1+y'^2}$ is the area (divided by 2π) of the surface made by spinning $y(t)$ about the t -axis. Two catenaries $y = c \cosh(t/c)$ pass through the ends. Predict which is the minimiser. Check all four views for both.

Checkpoint

$c = 0.848338$ (shallow) and 0.235095 (deep), from $c \cosh(1/2c) = 1$. Shallow: $y'(a) = -0.624109$, $J = 0.953624$, no conjugate point, $\lambda_1 = +5.839$, $\Delta J > 0$. Deep: $y'(a) = -4.134382$, $J = 1.089520$, $t_c = 0.15291144$, $\lambda_1 = -2.366$, $\Delta J < 0$. The conjugate point agrees to eight digits with Lindelöf's classical construction — the tangents to the catenary at a and at t_c meet the t -axis at the same point [4] — which uses no Jacobi equation at all.

Experiment 5: the fold

Stay on the minimal surface and widen the gap symmetrically: set $a = -L$, $b = L$ with $L = 0.6, 0.65, 0.66$, then 0.67 . Watch the deep catenary's t_c approach b , the two roots of $r(p)$ approach each other, and then disappear.

Checkpoint

Two catenaries exist only for $L < L^* = 0.662743$ (where $x \tanh x = 1$, $L^* = x / \cosh x$). The gap $b - t_c$ on the deep one: $0.347, 0.307, 0.180, 0.092$ at $L = 0.5, 0.6, 0.65, 0.66$. At $L = 0.67$ there is no extremal at all: the soap film breaks. At the fold, b is conjugate to a and $r'(p) = h(b) = 0$ — the two extremals merge there.

Experiment 6: never a conjugate point

Preset $F = y'^2 + y^2$, then *Arc length*. Predict h for each.

Checkpoint

$y'^2 + y^2$: $h = \sinh t$, never zero; arc length: $h = t - a$, never zero. Both are minimisers on every interval.

Experiment 7: your own Lagrangian

Choose *Custom*. Type $F = y'^2 - 4y^2$ on $[0, 3]$ with $y(0) = 0$, $y(3) = 1$. Predict t_c before pressing *Solve*. Then try $y'^2 - y'^2$: what does the Legendre line say, and why is the Morse count marked not applicable?

Checkpoint

$y'' = -4y$, $h = \frac{1}{2} \sin 2t$, $t_c = \pi/2 = 1.570796$, not a minimiser. For $F = y^2 - y'^2$, $P = -2 < 0$: Legendre's necessary condition fails, so the extremal is not a minimiser whatever the Jacobi field does; fast wiggles make $\delta^2 J$ unboundedly negative, so every fine mode is negative.

8 How the code is organised, and how it is checked

file	role
<code>cov_problem.m</code>	text $F \rightarrow g, g_y, g_{y'}, P, R, Q_0$ (symbolic)
<code>cov_shoot.m</code>	extremal + Jacobi field from (a, y_a) with slope p ; zeros of h
<code>cov_extremals.m</code>	scan $r(p)$, refine sign changes: every extremal in the range
<code>cov_perturbed.m</code>	the neighbour with slope $p + \delta$ and its crossings
<code>cov_second_variation.m</code>	finite-element $\delta^2 J$: eigenvalues, η^* , exact ΔJ
<code>cov_functional.m</code>	$J[y_0 + \eta]$ by 4-point Gauss quadrature
<code>conjugate_point_explorer.m</code>	the GUI (programmatic, with a handle API for tests)
<code>conjugate_point_study.m</code>	every fact in this guide, computed and checked
<code>run_conjugate_tests.m</code>	the suite, one exit code

Nothing is checked against a copy of itself. The oracles are closed forms (the oscillator's $\sin t$, $\cot b$, $k\pi$ and $2((k\pi/b)^2 - 1)$; the catenary's $c \cosh(1/2c) = 1$ and its area), a geometric construction (Lindelöf's tangents), and a second instrument (the Morse count against the Jacobi zeros, and $r'(p)$ by finite differences against $h(b)$). A mutation test corrupts the Jacobi equation, the Legendre coefficient and the integrand in turn, and requires the check aimed at each corruption to fail — and only that one.

9 The same idea in the orbit-transfer library

The conjugate-point test in the costate library (`orbit_transfer/costate_common/ms_conjugate_test.m`, `conj_spectrum.m`) is this construction in six dimensions. The “initial slope” is the initial costate $\lambda(0)$; the Jacobi field is the block of the state-transition matrix $\partial(r, v)(t)/\partial\lambda(0)$, restricted by a projection P to the costate directions that keep the problem normalised; and the conjugate point is where that block, augmented by the flow direction f_{rv} to account for the free final time, loses rank: $\det[\Phi_{rv}P, f_{rv}] = 0$ — the multidimensional $h(c) = 0$. A family of neighbouring min-time trajectories, all leaving the same departure state with slightly different costates, refocusing before arrival: the same picture as the orange curves in Figure 1.

References

- [1] I. M. Gelfand and S. V. Fomin, *Calculus of Variations*, Prentice-Hall, 1963, Ch. 5 (the second variation; Jacobi's equation; conjugate points; sufficient conditions for a weak extremum).
- [2] D. Liberzon, *Calculus of Variations and Optimal Control Theory: A Concise Introduction*, Princeton University Press, 2012, Ch. 2.

- [3] J. Milnor, *Morse Theory*, Annals of Mathematics Studies 51, Princeton University Press, 1963, Part III (the index theorem).
- [4] G. A. Bliss, *Calculus of Variations*, Carus Mathematical Monographs No. 1, Mathematical Association of America, 1925 (the surface of revolution of minimum area).